Table 2: **Split reconstruction across selection rules and component ablations.** The reserved cohort has 28 parents, 1,120 Gaussian cases and 448 corrupted cases. RMSD is mean per-case error in mÅ. Joint correctness requires both original and changed labels to be correct; false restoration assigns the changed case its parent class. Joint/restoration denominators are 420/184 and original-control denominators 280/224 (Gaussian/corrupted). One-SE is development-selected primary; fixed $k = 4$ is prespecified secondary. Component-removal rows retain their recorded selection rules. †The retrospective no-refit ablation includes changes in fold losses and selection. Bold identifies full-method variants.

	Gaussian noise				One corrupted observation			
Configuration	RMSD ↓	Joint ↑ (%)	Restored ↓ (%)	Original ↑ (%)	RMSD ↓	Joint ↑ (%)	Restored ↓ (%)	Original ↑ (%)
Huber triangulation (baseline)	5.271	75.7	9.5	92.9	7.395	85.9	0.0	88.8
Without physical departures ($D = 0$)	3.921	81.7	11.0	96.4	5.026	93.5	0.0	96.4
Without observation errors ($E = 0$)	3.377	83.3	0.0	87.9	8.757	37.0	0.0	46.4
Without joint refit, one-SE†	3.486	76.0	0.0	82.1	5.888	90.8	0.0	92.4
Split, minimum loss	3.387	83.3	0.0	87.9	4.421	85.9	0.0	88.8
Split, one-SE (primary)	3.336	86.2	0.2	90.7	4.349	90.8	0.0	92.9
Split, fixed $k = 4$ (secondary)	3.325	88.3	0.7	93.6	4.348	93.5	0.0	95.5